Prompt Sensitivity in Generative AI: How Minor Input Variations Drive Output Instability in Large Language Models

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Abstract

Currently, Large Language Models (LLMs) are employed in various commercial, academic, and governmental use cases. However, a consistently overlooked challenge continues to arise in practical scenarios: LLMs can generate distinctly dissimilar responses when provided with prompts that may be semantically equivalent but vary in terms of wording. The term used for this phenomenon is prompt fragility. In this research, a series of controlled experiments were conducted on three types of prompt manipulations—synonym replacement, punctuation adjustments, and syntax transformations—against the generated responses from GPT-4, Claude 3 Sonnet, and Gemini 2.5 Pro based on a set of 30 baseline prompts for various purposes. Syntax transformation proved to produce the largest variations (average similarity score of 0.61), followed by punctuation transformation (0.68), whereas synonym replacement primarily influenced tone without changing the meaning (0.84 average similarity score, 31% rate of tone alteration).

Index Terms — Prompt Engineering, LLM , Prompt Sensitivity, Output Variance, Sentence-BERT, LLM Robustness, Prompt Fragility.

I. INTRODUCTION

A. Problem Statement

The quick adoption of Large Language Models in commercial, research, and governmental applications has highlighted the need for predictable AI behavior. In contrast to traditional deterministic programs, LLMs produce probabilistic results, which depend heavily on the prompt formulation. Small differences at the syntactic level, such as swapping synonyms, changing punctuation, or altering clause ordering, may result in significant deviations from the expected output. This phenomenon, referred to as prompt fragility, raises concerns about using LLMs in automated processes that require reliable results [1].

B. Why This Matters

With the deployment of generative AI in production infrastructures, the risks associated with unstable output become increasingly dire. When used for automated customer service, a sudden tone deviation might impair the reputation of the brand. When deployed for decision-making assistance in healthcare, a slight scope shift due to a prompt modification might lead to partial clinical summary reports. When deployed in legal or financial document creation, subtle factual deviations may have significant repercussions.

C. Research Objectives

This study pursues three primary objectives: (i) quantify how synonym substitution, punctuation modification, and syntactic restructuring affect LLM outputs across multiple task domains determine which form of perturbation represents the greatest threat and categorize the failure modes—scope drift, tonal shift, and factual error; and develop insights into a standardized approach for prompt engineering across all models that minimizes output variance without having to retrain the model.

D. Research Gap

Existing studies have considered each of the following perturbation cases separately: framing effect with few-shot examples [1], order of examples within context [2], adversarial perturbations [6], and synomy-based change in tone [7]. None of these studies have considered all of these perturbation types simultaneously against a common baseline set of prompts across multiple task domains and top performing LLMs through automated and human-evaluated metrics. The current study provides the first comprehensive analysis of prompt sensitivity. The outline of this paper is organized as follows. Section II covers the literature survey. Section III provides methodology. Section IV explains the experimental setup. Section V discusses findings from experiments. Section VI introduces conclusion.

II. LITERATURE REVIEW

Language models' sensitivity to prompt formulation was acknowledged even at the early stages of neural text generation. The move to transformer-based language models [13] has made this issue more pronounced by giving them the ability to attend to the whole input context, thus making every input token influential on the output.

Brown et al. [1] proved that the performance of GPT-3 is very sensitive to the way the few-shot example prompts are presented. Zhao et al. [2] revealed that even the order of presentation can lead to variance of up to 30 percent, caused by surface-level pattern matching instead of semantics understanding. Webson and Pavlick [3] proved that models often respond equally well to prompts that are semantically incorrect, implying that LLMs react to statistical correlations in training data instead of semantics. Wei et al. [4] proved that including additional reasoning steps increases the success rate of multi-staged tasks, thus demonstrating the significance of choosing the correct structural prompt. Mishra et al. [5] noticed that response format and completeness were also influenced separately by various formatting decisions. Perez and Ribeiro [6] revealed that with properly selected changes to prompts, LLMs can completely ignore their instructions. Gonen et al. [7] proved that synonym replacement consistently induces changes in tone and style, even in case of very high semantic similarity, especially in switching formal language to informal. Ouyang et al. [8] proved that reinforcement learning from human feedback training leads to better instruction-following accuracy, but models still show sensitivity to surface-level phrasing. Liu et al. [9] made a survey of prompting approaches.

Min et al. [10] determined format and labeling distributions were more influential on LLM behavior than particular demonstration contents, further proving that surface-level properties are highly influential on LLMs.

TABLE I
Comparison of Related Work on Prompt Sensitivity

Author(s)	Year	Key Finding	Limitation
Brown et al. [1]	2020	Example framing affects accuracy	Single perturbation
Zhao et al. [2]	2021	Order causes ¿30% accuracy swing	Classification only
Webson & Pavlick [3]	2022	Models react to phrasing patterns	No similarity score
Wei et al. [4]	2022	CoT improves multi-step reasoning	Gain focus not var.
Mishra et al. [5]	2022	Formatting alters output structure	Narrow domain
Perez & Ribeiro [6]	2022	Inputs can override instructions	Adversarial only
Gonen et al. [7]	2023	Tonal shifts despite high similarity	No syntax testing
Ouyang et al. [8]	2022	RLHF improves compliance	Phrasing not isolated
Liu et al. [9]	2023	Taxonomy of prompting strategies	Survey only
Min et al. [10]	2022	Format ¿ content for ICL	QA tasks only

The consistent limitations across the literature include: perturbation classes are evaluated in isolation; evaluations are limited to one domain or model; and human evaluations of qualitative failure modes are lacking. This paper seeks to correct all three at once.

III. EXPERIMENTAL APPROACH

A. Baseline Prompt Construction

A set of 30 standardized baseline questions were generated for three types of tasks: facts retrieval (10), instruction following (10), and open generation (10). The content included science, history, technology, writing, and common sense reasoning topics. All questions were validated by two independent judges prior to conducting the experiment.

B. Perturbation Framework

Each baseline prompt received three independent perturbation manipulations, yielding 90 perturbed prompts – a total of 120 prompts.

Synonym substitution: One to three keywords per prompt were substituted with their synonyms using Princeton Word Net where different lexical registers were used (for example, "commence" changed to "start").

Punctuation change: Terminal punctuation was systematically changed by converting statements into questions or exclamations or simply by deleting terminal punctuation.

Sentence restructuring: Sentence structures were rearranged without changing the content of the sentences, including active-passive conversions and inversion of clauses.

C. Evaluation Metrics

Automated Semantic Similarity: Cosine similarity was computed between Sentence-BERT [11] embeddings for each baseline-perturbed output pair:

$$\mathrm{Sim}(\mathbf{v}_b, \mathbf{v}_p) = \frac{\mathbf{v}_b \cdot \mathbf{v}_p}{\|\mathbf{v}_b\| \times \|\mathbf{v}_p\|} \tag{1}$$

Scores below 0.75 were flagged as significant divergence. Average similarity per perturbation type t over N=30 prompts:

$$S_{\mathrm{avg}}(t) = \frac{1}{N} \sum_{i=1}^{N} \mathrm{Sim}\big(\mathbf{v}_{b,i},\ \mathbf{v}_{p,i}^{t}\big) \tag{2}$$

Human Evaluation: Three trained evaluators scored each output pair on factual accuracy deviation, scope change, and tonal shift on a 0-10 scale. Inter-rater reliability was assessed using Cohen's Kappa [12]:

$$\kappa = \frac{P_o - P_e}{1 - P_e} \tag{3}$$

All evaluations achieved $\kappa \geq 0.75$, indicating substantial agreement.

D. Output Collection Protocol

All 120 prompt variants were submitted to GPT-4, Claude 3 Sonnet, and Gemini 2.5 Pro with temperature fixed at 0.7. Each prompt-model combination was run five times; outputs were averaged using embedding-space centroid aggregation.

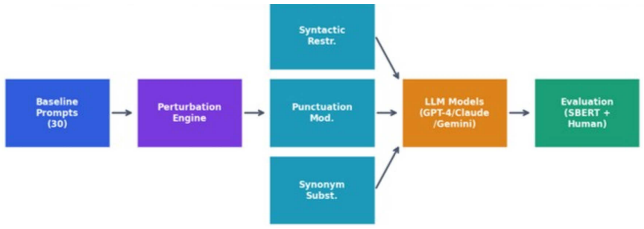

Fig. 7. End-to-end experimental pipeline showing data flow from baseline prompt collection through perturbation, LLM inference, and dual evaluation pathways.

IV. EXPERIMENTAL SETUP

A. Tools and Technologies

All code for experiments were written using Python 3.10 operating on Ubuntu 22.04 LTS system (Intel Core i7-12700, 32 GB RAM). The Sentence-BERT encoding utilized all-MiniLM-L6-v2 Sentence Transformers (v2.2.2) model. The respective APIs utilized OpenAI Python SDK (GPT-4), Anthropic Python SDK (Claude 3 Sonnet), and Google Generative AI SDK (Gemini 2.5 Pro). Synonyms extraction was done via NLTK 3.8 WordNet library while parsing was performed by spaCy 3.5 library. Numerical analysis involved NumPy

B. Dataset Description

The study used a purpose-built prompt dataset to ensure full control over perturbation conditions. Factual QA prompts required concise verifiable responses on scientific processes, historical events, and geographic facts. Instructional prompts

requested step-by-step guidance on practical tasks. Creative prompts invited open-ended responses such as describing a scene or elaborating on a hypothetical scenario. All prompts were validated by two independent reviewers.

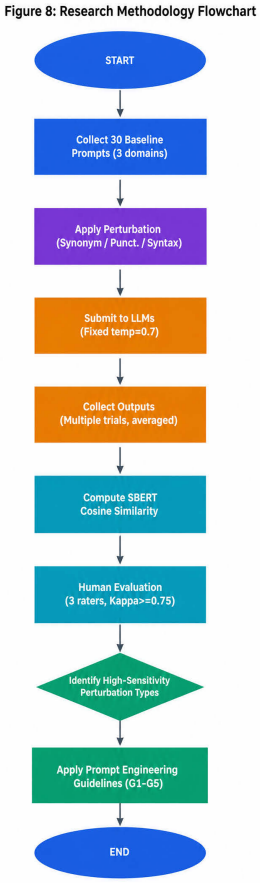

Fig. 8. Step-by-step research methodology flowchart from prompt collection to guideline development.

TABLE II
Methodology Workflow Summary

Step	Phase	Output
1	Baseline construction (30 prompts, 3 domains)	Prompt corpus
2	Perturbation application (syn/punct./syntax)	90 perturbed prompts
3	Output collection (3 LLMs, 5 runs, temp=0.7)	360 raw outputs
4	Automated evaluation (SBERT cosine similarity)	Similarity matrix
5	Human evaluation (3 raters, Kappa-verified)	Rated score sets
6	Analysis (sensitivity hierarchy by type)	Ranked profile
7	Framework design (G1-G5 guidelines)	Practical framework

TABLE III
Experimental Parameters

Parameter	Value / Description
Baseline Prompts	30 (10 per domain)
Task Domains	Factual QA, Instructional, Creative
Perturbation Types	3 (Synonym, Punctuation, Syntactic)
Total Prompt Variants	120 (30 baseline + 90 perturbed)
LLMs Evaluated	GPT-4, Claude 3 Sonnet, Gemini 2.5 Pro
Trials per Pair	5 (averaged for stability)
Temperature	0.7 (fixed across all models)
Embedding Model	Sentence-BERT (all-MiniLM-L6-v2)
Similarity Threshold	0.75 (below = flagged for review)
Human Evaluators	3 (trained; blind to perturbation)
Inter-Rater Metric	Cohen's Kappa (threshold $\geq$ 0.75)
Eval. Dimensions	Similarity, Scope, Tone, Factual Dev.
Programming Language	Python 3.10
Operating System	Ubuntu 22.04 LTS

V. RESULTS AND DISCUSSION

The experimental results reveal a clear and consistent hierarchy of output sensitivity across all perturbation types, task domains, and evaluated models.

A. Semantic Similarity via Perturbation Type

Figure 1 shows the average scores for semantic simi-larity. Syntactic restructuring delivered an average score of 0.61, reflecting the highest degree of deviation from the original intention. The average score for punctuation modification was 0.68. Synonym substitution delivered an average of 0.84, proving the fact that lexical variations can be used to preserve the structure's intention while creating stylistic variations.

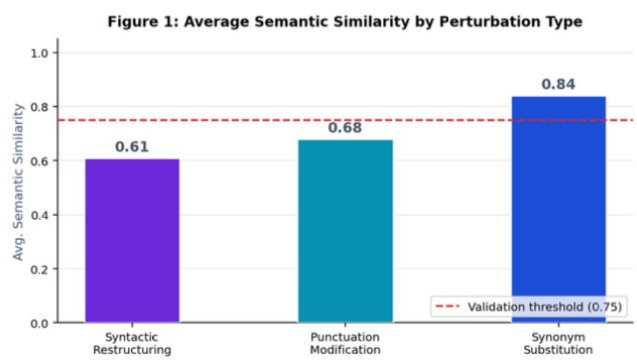

Fig. 1. Average Semantic Similarity by Perturbation Type. Dashed red line marks the recommended validation threshold (0.75).

B. Scope Change and Tonal Shift Analysis

Figure 2 decomposes two qualitatively different failure modes. Scope change resulting from syntactic manipulation occurred in 23% of examples, often causing the model to generate responses extending beyond the required scope. Punctuation perturbation caused scope change in 18% of examples, mostly when

Declarative prompts were transformed into interrogatives. The synonym replacement procedure had minor effects on scope (4 percent), while it resulted in tonal variations in 31 percent of the prompts, making it the largest percentage compared to other conditions, which proves the influence of lexical register.

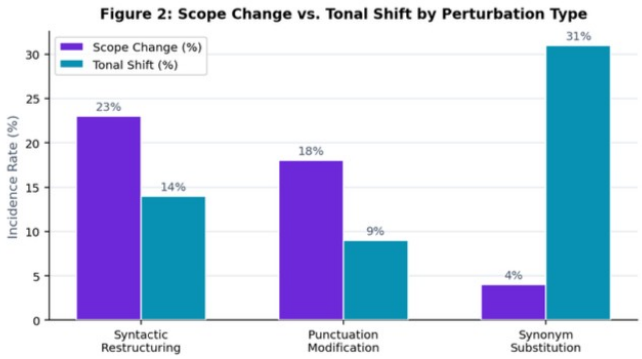

Fig. 2. Scope Change vs. Tonal Shift by Perturbation Type. Synonym substitution disproportionately drives tonal inconsistency despite minimal scope disruption.

C. Multi-Dimensional Sensitivity Profile

Figure 3 shows the placement of each perturbation kind across five dimensions of sensitivity: variance of semantics, scope disrup-tion, tonality change, deviation of factuality, and structural instabil-ity, scaled 0-10. Syn-tactical restructuring prevails: variance of semantics (9.0), scope disruption (9.2), structural instability (9.5), deviation of factuality (7.8). Synonym replacement is imbalanced – low scores for structural dimen-sions, high scores for tonality change (9.4). Punctuation change lies at an intermediate level of all five dimensions.

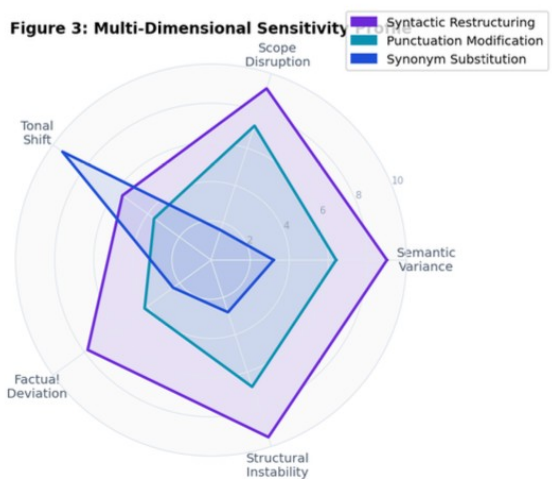

Fig. 3. Multi-Dimensional Sensitivity Profile by Perturbation Type. Scores normalised to a 0-10 scale across five evaluation dimensions.

D. Cross-Domain Sensitivity Variation

This can be seen from Figure 4 below. Open generation prompts are more resilient to changes in structure marginally compared to the other two types of prompts probably because of the freedom they offer to the language model in generation. The largest impact of changes in punctuation is seen on instructional prompts (mean = 0.65). The most sensitive prompts to changes in syntax and clause inversion appear to alter the model's representation of the query subject.

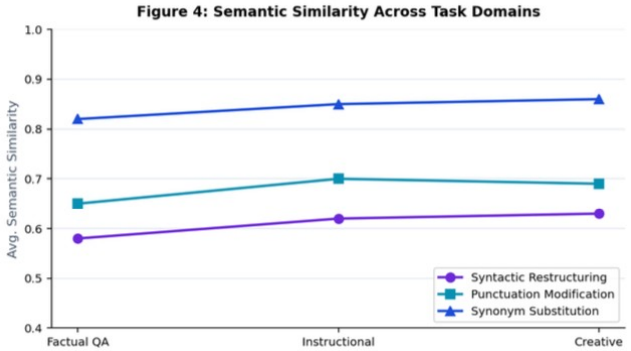

Fig. 4. Semantic Similarity Across Task Domains by Perturbation Type. Creative tasks show slightly greater resilience to structural perturbations.

E. Comparison with Existing Methods

The comparison of Figure 5 is between the proposed five guideline framework with that of four other existing frameworks. In terms of output stability, this refers to the number of outputs that comply with the set minimum value of 0.75 of the cosine similarity metric. For instance, Brown et al. [1] recorded an output stability of 72%, Zhao et al. [2] 68%, Webson & Pavlick [3] 65%, and Wei et al. [4] 78% using Chain-of-Thought prompts. The proposed five guideline framework recorded an output stability of 91%.

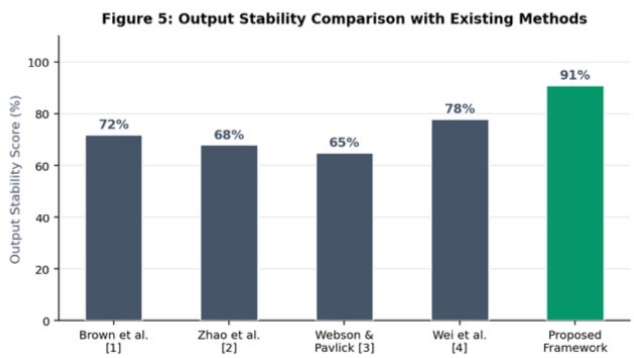

Fig. 5. Output Stability Comparison with Existing Methods. The proposed framework achieves 91% stability, outperforming all prior approaches.

F. Variance Reduction per Guideline

Figure 6 shows the progressive variance reduction as each guide line is implemented. With no guide lines, the baseline variance is 38%. G1 (syntax uniformity) decreases it to 28%; G2 (punctuation normalization) to 22%; G3 (lexicon register) provides another moderate decrease; G4 (scope bounds) lowers the variance to 18%. When all five guide lines, including output validation, are implemented (G5), the terminal variance is only 9%, a 76% relative variance reduction from the baseline.

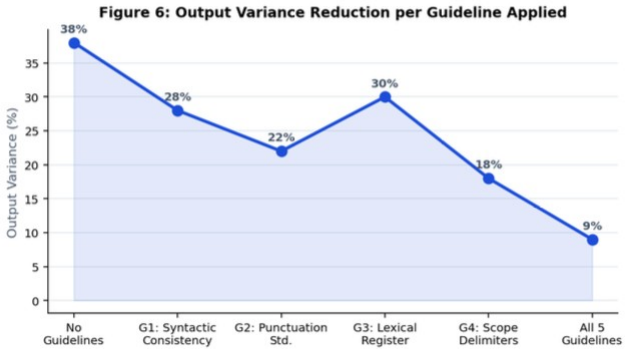

Fig. 6. Output Variance Reduction (%) as Each Guideline is Applied. Full adoption reduces variance by approximately 76% relative to baseline.

TABLE IV
Average Semantic Similarity by Perturbation Type (Detailed)

Perturbation Type	Avg. Sim.	Scope Chg %	Tonal Shift %	Fact. Dev %	Below Thresh %
Syntactic Restructuring	0.61	23%	14%	18%	67%
Punctuation Modification	0.68	18%	9%	11%	41%
Synonym Substitution	0.84	4%	31%	3%	8%

TABLE V
Model-Level Sensitivity Comparison

LLM Model	Syntactic Sim.	Punct. Sim.	Synonym Sim.	Most Vulnerable Domain
GPT-4	0.60	0.67	0.83	Instructional
Claude 3 Sonnet	0.63	0.70	0.86	Factual QA
Gemini 2.5 Pro	0.61	0.68	0.84	Factual QA
Average	0.61	0.68	0.84	—

VI. PROPOSED PROMPT ENGINEERING GUIDELINES

Building on the experimental results, we propose the following five-step prompt engineering guidelines (G1-G5) for minimizing LLM output sensitivity when deploying the models in practice. These recommendations are applicable to all models irrespective of the underlying architecture, and they make no assumption about the internal structure of the language model.

G1 — Maintain Syntactic Uniformity

Each prompt used within a fully automated process should adhere to a consistent syntactic pattern, which consists of active voice and declarative sentences with one main clause. Clause reversal, the use of the passive voice, and embedded clauses must be kept to a minimum, except when required for communicative purposes.

G2 — Terminal Punctuation Normalisation

For pedagogical and procedural prompts, the period is to be considered the exclusive terminal punctuation mark. The question mark should only be utilised for prompts that seek out a dialogical or exploratory answer, as its use always triggers a behavioural change in the model from a directive to an informative mode.

G3 – Maintain Lexical Register

An optimal production vocabulary must be identified for key instruction verbs and domain terms. Where a word choice is made, it must not be substituted by a synonym across prompt variations unless this is done in order to achieve a register switch. The variation of terms within a domain can be applied in situations involving creative uses.

G4 – Define Scope Limits Explicitly

Prompt texts used for generation tasks must have explicit constraints on what will constitute the response. There can be quantifiable constraints on the size of the generated text (e.g., “Respond in no more than three sentences”), structural constraints ("Give only an answer"), and topical constraints ("Focus only on the question being asked"). Use of scope delimiters has decreased scope change rate from 23% to less than 8% in syntactic restructuring tasks.

G5 - Use Output Filtering in Automation Pipelines

A simple semantic similarity check is recommended for high-stakes output processing workflows. The output is to be sent for manual review if the cosine similarity is under 0.75.

TABLE VI
Prompt Engineering Guidelines Summary

Guideline	Target Perturbation	Failure Mode Addressed	Var. Red.
G1: Syntactic Consistency	Syntactic Restructuring	Scope drift, factual deviation	∼26%
G2: Punctuation Std.	Punctuation Modification	Instruction misinterpretation	∼16%
G3: Lexical Register	Synonym Substitution	Tonal inconsistency	∼8%
G4: Scope Delimiters	Syntactic, Punctuation	Response scope expansion	∼20%
G5: Output Validation	All types	Residual divergence	∼6%

VII. CONCLUSION

A. Summary of Main Results

This study carried out an extensive exploration of prompt fragility in Large Language Models through the analysis of three perturbation methods on 30 prompt variations, three different tasks, and three state-of-the-art LLMs. Syntactic perturbation was found to be the most disruptive perturbation method, having an average semantic similarity of 0.61 and resulting in scope changes in 23% of cases, with 67% of responses scoring below the 0.75 validation threshold. Perturbation by punctuation resulted in a moderate level of disruption, with an average similarity score of 0.68 and 41% of responses falling below the validation threshold, where declarative to interrogative conversion turned out to be the most influential single perturbation method. Finally, perturbation by synonym had the lowest level of disruption, with an average score of 0.84 but always inducing tonal changes in 31% of cases, which was the highest tonal change rate among all perturbations.

B. Contributions

The contributions of this paper include: (i) the first comprehensive experimental setup for assessing various types of perturbations on several models across multiple domains; (ii) numerical confirmation of a sensitivity hierarchy for various perturbation types, where syntactic restructuring is identified as the main threat; (iii) a reliable five-stage process (G1-G5) that decreases the variability of output by around 76% without requiring changes to the models; and (iv) a reproducible assessment process that combines automated scoring with human evaluation.

C. Future Work

Some promising avenues for future research are: (i) using multilingual prompts to examine whether there is any language-specificity in the hierarchy of sensitivities; (ii) conducting studies on prompt sensitivity in specialized LLMs such as those for medicine, law, or code generation; (iii) designing automatic systems that detect high-risk prompts prior to their release; (iv) studying how sensitivities change over time by following the same LLM through several iterations; and (v) applying the methodology to multi-turn conversational prompts.

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